

Special Article

Quality and safety considerations in stereotactic radiosurgery and stereotactic body radiation therapy: Executive summary

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Introduction

This executive summary briefly describes the overall goals and content of the report on safety consideration for stereotactic radiosurgery (SRS) and stereotactic body radiation therapy (SBRT), but is very limited in length and content. This abridged version is not intended to replace the full length report but rather to highlight key recommendations. All readers are referred to the full report published online only at www.practicalradonc.org.

This report on SRS and SBRT is part of a series of white papers addressing patient safety commissioned by the American Society for Radiation Oncology (ASTRO). This document was approved by the ASTRO Board of Directors on April 11, 2011. It has been endorsed by the American Association of Physicists in Medicine, the American Society of Radiologic Technologists, and the American Association of Medical Dosimetrists. It has been reviewed and accepted by the American College of Radiology's Commission on Radiation Oncology.

In addition to many academic papers, professional organizations in North America have previously published several “guidance” reports on various aspects of SRS/SBRT.¹⁻⁶ Several recent national and international efforts that specifically address safety in radiotherapy also played a prominent role in formulating the recommendations in this report.⁷⁻¹¹

It is important to understand that the SRS/SBRT measures described and recommended in this document are just one component of a broader process of ongoing quality assurance (QA) that includes periodic review of errors, incidents, and near misses for the purpose of developing or refining standard operating procedures that minimize the risk of such events. Similarly, detailed equipment specifications and tolerances have been described in a number of documents, and while some of these aspects may be reiterated and emphasized in this paper, it is not intended to be comprehensive in this regard. Rather, this report builds on these and other documents, broadly addressing SRS/SBRT delivery with a primary focus on programmatic elements and human processes that can identify and correct potential sources of error, particularly those which can result in catastrophic consequences.

SRS has been used for decades in the treatment of brain metastases and a variety of other cranial neoplasms and functional disorders; its efficacy and toxicity profile have been well described and its role well established as an efficient and effective means of achieving a high rate of local control and, in some settings, improved survival.¹² Acute side effects, including headache, pin-site infection, and short-term exacerbation of neurologic symptoms are relatively minor and readily managed. Late side effects, including radiation necrosis, brain edema, and exacerbation of preexisting (or development of new) neurologic deficits occur in less than 5% of patients.¹³ Five-year local control rates following SRS or FSRT [fractionated stereotactic radiation therapy] for acoustic neuromas exceed 95%.¹⁴ Current doses of 13 Gy (single fraction) or ~50 Gy (in 1.8-Gy fractions) yield excellent rates of hearing preservation and very low rates of facial and trigeminal neuropathies. Similarly, excellent rates of local control can be expected following either SRS or FSRT treatment of meningiomas.¹⁵⁻¹⁷

SBRT is a much more recent modality, with unique technological and clinical considerations. Nevertheless, initial clinical results from prospective single institution, and more recently, multi-institutional clinical trials of SBRT have documented similar high rates of tumor control coupled with a low incidence of serious toxicity despite the high dose fractions of radiation being delivered. The efficacy of SBRT is established for a variety of clinical indications as a primary treatment for selected early-stage cancers or as treatment for discrete tumors in patients with oligometastatic disease, selected benign neoplasms in or near the central nervous system, or

recurrent cancer in previously irradiated regions. The utility of SBRT is perhaps best exemplified in the case of inoperable early-stage lung cancer,¹⁸ where the 3-year primary tumor control rate of 98% is roughly twice what would be expected from conventional RT given over a 6- to 7-week period. To date, reports of prospective clinical trials of SBRT have typically documented similar high rates of tumor control, coupled with a low incidence of serious toxicity, despite the high-dose fractions of radiation given to tumors.¹⁹⁻²⁹ This favorable therapeutic ratio is achieved because SBRT couples a high degree of anatomic targeting accuracy and reproducibility with very high doses of precisely delivered radiation, thereby maximizing the cell-killing effect on the target(s) while minimizing radiation-related injury in adjacent normal tissues.

Given that very high-dose fractions of radiation are delivered, the margin of error for SRS and SBRT is significantly smaller than that of conventional radiotherapy and therefore special attention and diligence is required. A small error in target localization for any 1 fraction risks undertreatment of portions of the tumor by 20% or more, and inadvertent overdosage of adjacent normal tissues could escalate the risk of serious injury to a much greater degree than an equivalent treatment error in a course of radiotherapy where a substantially lower dose per fraction is used.^{25,30-34}

Many in the community are aware of recent events, publicized in the media, in which serious errors have occurred. These include the following: a calibration error on a radiosurgery linac that affected 77 patients in Florida in 2004-2005; similar errors in measurement of output factors affecting 145 patients in Toulouse, France in 2006-2007,³⁵⁻³⁷ and 152 patients in Springfield, MO from 2004 to 2009; an error in a cranial localization accessory that affected 7 centers in the U.S. and Europe; and errors in failure to properly set backup jaws for treatments using small circular collimators affecting a single arteriovenous malformation patient at an institution in France,³⁶ and 3 patients at an institution in Evanston, IL.³⁸

While no side effects related to the Florida calibration error have been reported, that is not the case with several of the other events. Gourmelon et al reported a 31% 12-month actuarial rate of trigeminal neuropathy in 32 acoustic neuroma patients overdosed in the Toulouse accident.³⁷ In contrast, despite a mean overdose of 61.2%, no treatment-related morbidity was observed in the 33 patients treated for brain metastases.³⁵ The French patient treated with the incorrect backup collimator setting developed an oeso-tracheal fistula requiring surgery, experienced a hemorrhage, and subsequently died.³⁶ One of the 3 Evanston patients, treated for trigeminal neuralgia, is described as being in a vegetative state.³⁸

Further, radiosurgery errors are not limited to any particular technology. As an example, challenges in accurate measurement of output factors such as those encountered on linacs in Toulouse, France and Springfield,

MO have also been encountered on gamma devices. In 1998, the output factor for a 4-mm gamma collimator was corrected by approximately 10%, from 0.80 to 0.87, by the manufacturer.³⁹⁻⁴⁰ A review of the Nuclear Regulatory Commission Radiation Event Notification Report database yielded 13 gamma-based radiosurgery-related events from 2005 to present, 12 of which resulted in a deviation from the original prescription. Seven of the events involved the treatment of the wrong location, while 3 events involved delivery of an incorrect dose. While patient outcome is not described on the Nuclear Regulatory Commission site, several of the events listed, including treatment of the wrong location with single fraction doses as high as 90 Gy, would likely be accompanied by significant morbidity. Wrong-site errors continue to plague all medical disciplines and are not unique to radiotherapy.⁴¹

The accidents described can largely be attributed to human error, mirroring the radiotherapy experience throughout the United Kingdom, in which only 2 out of 181 incidents reported since 2000 were determined to be nonrelated to human error.⁸ However, other factors also contributed. These include limits in equipment safety design and the inadequacy of systems and procedures to ensure that the stereotactic treatment was robust to the sources of error that eventually contributed to failure. Clearly then, improvement in human knowledge, training standards, and implementation of robust QA processes are needed to minimize these errors, which in the case of SRS and SBRT, can have catastrophic consequences. Recommendations designed to guard against catastrophic failure in SRS and SBRT are provided in Appendix 1 of the online document.

Fundamental elements of SRS and SBRT safety and quality

It is important to emphasize that SRS and SBRT are not 1 treatment technique or modality. The implementation and accompanying requirements for immobilization, simulation, treatment planning, delivery, and quality assurance can vary significantly with disease site. Clinical and technical proficiency for one site (eg, spine) does not always translate to proficiency in another site (eg, lung). This complex nature of the stereotactic treatment process, and the consequences of errors when delivering high-dose fractions of radiation, mandates a systematic and prospective approach to each disease site. Many of the overall recommendations of the 2008 document *Towards Safer Radiotherapy*⁸ are appropriate for SRS and SBRT programmatic development, including the following: a multidisciplinary working environment with a culture that fosters clear communication and guards against inappropriate interruptions; careful planning and thorough risk assessment when introducing new techniques and tech-

nologies; a thorough review of all resources including staffing levels and skills; thorough training of all personnel, to include training in quality management and safety practices in addition to program-specific education; development of quality assurance processes that encompass all clinical and technical program aspects; and development of checklists, processes for documentation and reporting, peer review, regular review of processes and procedures, updating of clinical guidelines and recommendations, ongoing needs assessment, and continuous quality improvement.

Personnel considerations

SRS and SBRT require a large commitment of resources. Personnel resources required for proper operation of an SBRT program can therefore be expected to be significantly greater than for a traditional radiation therapy program.^{6,42} Further, SRS and SBRT require the coordinated efforts of a team of properly trained individuals who assume essential roles during the patient evaluation and treatment process.⁴⁻⁸ In addition to clinic nurses and other staff who provide general support for all patients, the essential personnel for SRS/SBRT include radiation oncologists, medical physicists, dosimetrists, and radiation therapists. Other physicians may participate in the care of patients undergoing SRS or SBRT by offering assistance derived from their own subspecialty training and expertise; examples include neurosurgeons, pulmonologists, hepatologists, and oncologic surgeons. Because the resources required to implement and manage an SRS/SBRT program are significant, institutional administrators must be part of the overall team. All program personnel must demonstrate initial attainment of knowledge and competence in their respective discipline through graduation from an approved educational program, board certification, and licensure as appropriate. Training on SRS/SBRT technologies and their specific clinical application, including training provided by the equipment vendor(s), is an essential program element; all program individuals must receive initial SRS/SBRT-specific training for each disease site, and must participate in SRS/SBRT-specific continuing medical education.

Treatment of various disease sites should be considered within the context of nationally accepted clinical standards. Based on program goals and patient selection criteria, it is likely that treatment guidelines and procedures will be site-specific. Prior to initiating an SRS or SBRT program, this report strongly recommends that plans for patient selection and treatment guidelines be developed and clearly documented within each institution.

Technical considerations

SRS and SBRT require the use of technology at a standard above that routinely considered minimally

acceptable for conformal radiotherapy and initial image-guided radiotherapy applications. The extreme demands imposed by the ablative paradigm of dose delivery amplify concerns over the volume of tissue irradiated to high doses as well as doses in serial organs and regions near the skin that may otherwise be ignored. To achieve these demands, small margins around the clinical target volume are necessary to such an extent that conventional radiographic localization based on bony anatomy is generally insufficient. A comprehensive image guidance and motion management strategy needs to be applied and maintained with sufficient technology and procedures to ensure safe and effective positioning for treatment. Furthermore, the dose distributions considered acceptable for SRS and SBRT require the use of large numbers of non-opposing beams, often inclusive of multiple non-axial approaches, to achieve the dosimetric goal of confining the high-dose region to the volume of interest while effectively minimizing peripheral dose.^{43,44} Due to needs of clearance for beam angles and imaging, isocenter placement may be nontraditional. Dose needs to be calculated accurately through complex heterogeneities and represented over the entire irradiated volume.

SRS/SBRT begins to deviate from conventional treatments at simulation. Typical immobilization equipment for SBRT includes custom-formed devices that cover a large extent of the patient above and below the tumor (eg, evacuated bean bags). The use of ancillary localization and position monitoring technologies, such as surface imaging techniques, implanted radiographic markers, and electromagnetic transponders, may play a role in specific disease sites. For each of these devices and indications for use, the operational team (radiotherapy technologist, MD, physicist) should establish procedures for assessing the residual positioning uncertainty that is possible when combining these immobilization means with specific image guidance strategies. Imaging needs for simulation and planning may include detailed motion estimation (eg, 4-dimensional computed tomography [4DCT]), as well as inclusion of enhanced soft tissue (magnetic resonance imaging), or metabolic (positron emission) information. Paraspinal SBRT may require enhanced visualization of the spinal cord (eg, through MR or CT myelography).

The treatment planning environment must be capable of supporting multimodality and multidimensional input data for SRS and SBRT planning. Specifically, magnetic resonance imaging, positron emission tomography, and multiple CT scans (eg, non-contrast and contrast, 4D) must be able to be combined in an accurate manner to facilitate target and normal tissue definition, to establish a patient data set for use in image guidance, and to generate an appropriate density grid for dose calculation. The planning system must be able to support dose calculation algorithms that represent dose deposition in the face of heterogeneities with sufficient accuracy.

Commercial planning systems using pencil beam algorithms generally do not meet this requirement. Demonstration of calculation accuracy during the commissioning process through an independent dosimetric check of a planned and irradiated phantom containing heterogeneities, by an independent entity such as the Radiological Physics Center, is strongly recommended prior to initiating an SBRT program.

Image-guided localization is increasingly used in SRS (ie, frameless radiosurgery) and is a prerequisite for all SBRT applications. A comprehensive image guidance and motion management strategy, therefore, needs to be applied and maintained with sufficient technology and procedures to ensure safe and effective positioning for treatment. Ideally, this guidance should involve tumor-based positioning at the start of each treatment fraction. In the absence of direct tumor localization, reliable soft tissue surrogates (eg, implanted fiducial markers) may be necessary as a means of estimating position. Conventional radiographic localization based on bony anatomy is generally insufficient to meet the precision demands of stereotactic treatments for soft tissue targets. Appropriate equipment for localization (eg, cone beam CT or other 3D image-based method) must be used and maintained with sufficient quality assurance procedures to ensure the usefulness (image quality) and accuracy of positioning. In addition to pre-treatment positioning, the management of intra-fraction patient body movement as well as physiological motions such as breathing must be accounted for. Some examples of such technologies include in-room surface monitoring systems, fluoroscopic observation, external gating systems, and external interventional mechanisms such as abdominal compression and active breathing control systems.

Acceptance and commissioning

Acceptance testing and commissioning are essential technical components of an SRS/SBRT program that must be performed and documented completely and thoroughly prior to clinical application. Acceptance testing is performed in cooperation with an equipment vendor to ensure that the equipment is operating within stated specifications and in compliance with regulatory requirements. As SRS/SBRT requires a high level of precision in target and dose localization, it is necessary for vendors to demonstrate that capabilities are commensurate with the requirements of SRS/SBRT.

Generally the commissioning task begins with the measurement of the radiation characteristics of a machine. Beam data acquisition is a common task performed routinely by medical physicists.⁴⁵ Acquisition of beam data for SRS and SBRT can be particularly challenging, however, due to the small size of the fields employed, and small field measurements require appropriately small detectors.^{4,6} Further, small photon beams

exhibit a loss of lateral electronic equilibrium on the central axis, producing output factors that fall off rapidly for fields below 10 mm in diameter.^{46,47} Due to the profound clinical consequences of incorrect beam data that are now well known in the recent media, this report strongly recommends an independent assessment of small field measurements. This could include comparison against published data, comparison against unpublished data from similar treatment units, or by verifying the data through a completely independent set of measurements. Additionally, independent verification of the absolute calibration, utilizing a service such as that provided by the Radiologic Physics Center, is essential.

Following beam data acquisition, the treatment planning system must be fully commissioned to ensure accurate calculation of dose and monitor units. This involves a systematic comparison of calculation and measurement ranging from simple configurations such as a single beam to sophisticated arrangements of beams encompassing any and all situations encountered in clinical practice.

Ultimately, acceptance testing and commissioning must be performed in a manner that assesses both the individual and integrated localization and dosimetric components in an end-to-end manner.⁴⁻⁶ This is stated very succinctly in the Canadian Association of Provincial Cancer Agencies stereotactic radiosurgery-radiotherapy standards [4]: "It is essential to recognize that commissioning SRS/T techniques involves more than just ensuring that the equipment itself works properly. The whole treatment chain, including the measuring, imaging modalities and treatment planning system must be tested in addition to the delivery unit and the SRS/T tools." This will most likely be facilitated by incorporating appropriate, site-specific anthropomorphic phantoms.

The Quality Assurance Program

Quality assurance is an essential aspect of every medical discipline, and the importance of a robust quality assurance program to reduce errors of all kinds cannot be overstated. In its Radiotherapy Risk Profile, the World Health Organization states that proper QA measures are imperative to reduce the likelihood of accidents and errors and increase the probability that the errors will be recognized and rectified if they do occur.⁷ ASTRO and ACR guidelines are equally clear with regard to SRS and SBRT QA: "Strict protocols for quality assurance must be followed."⁵ Additionally, as "the complexity, variation in individual practice patterns, and continued evolution of stereotactic-related technology can render a static, prescriptive QA paradigm insufficient over time,"⁶ QA activities must continually evolve. Programs must adhere to a process of ongoing quality improvement, continually evaluating the adequacy of policies and procedures. Each

of these elements is described in detail in the full version of this report (online only at www.practicalradonc.org).

Recommendations for stakeholders

While this report deals primarily with institutions and professional staff, there are many stakeholders in the safety/QA process, with common goals and shared responsibilities. In this regard, improvement of patient safety would be facilitated by collaborative efforts between the manufacturers and the users, specifically in designing safer systems, in developing QA methods and training programs, and in promoting patient safety for SRS and SBRT. Vendors must understand the needs and requirements of the clinicians, medical physicists and radiation therapists relative to the systems and processes for SRS and SBRT. With such understanding they must exert all the necessary efforts to incorporate features and safeguards to assure efficacious and safe operation of their products. Vendors must provide additional opportunities for specialized training, emphasizing clinical implementation and quality assurance in addition to technical aspects, and the home institution must make available resources and time for such training. Vendors must do more to emphasize all QA aspects, not only equipment QA, but process QA. SRS/SBRT systems consist of multiple components, and vendors must ensure and demonstrate full mechanical, electronic and information connectivity of these components. In situations where components or subsystems come from more than one manufacturer, it is the responsibilities of the manufacturers to collaboratively support compatibility of the various subsystems, and their safe operation when used in combination.

Professional organizations must also do more to facilitate proper training in specialized procedures such as SRS and SBRT, and to ensure that only qualified practitioners are involved in such procedures. Specialized accreditation programs may be an effective mechanism to realize this, and extending ACR specialty accreditation to SRS and SBRT would be a strong step in emphasizing and recognizing practice quality. The current ACR-ASTRO Radiation Oncology Accreditation Program should also become mandatory. Professional organizations must work closely with industry to enhance safety aspects of products and practices.

There are many steps that government agencies can take to enhance safety within the profession. There are numerous inconsistencies in regulation and radiation-event reporting between state and federal agencies, and with regard to radioactive versus X-ray sources. The findings of an earlier investigation on regulatory reform in radiation medicine pointed out several areas of need, yet many of the recommendations were never implemented.⁴⁸ Centralized registries for event reporting, such as mandated

by law in the United Kingdom, ensure appropriate transparency and provide an effective mechanism for all stakeholders to learn from mistakes. Several voluntary efforts currently exist, notably the Radiation Oncology Safety Information System⁴⁹ and the system implemented at Washington University.⁵⁰

Summary

In summary, SRS and SBRT require a team-based approach, staffed by appropriately trained and credentialed specialists. SRS and SBRT training should become a required part of radiation oncology residency training and of Accreditation of Medical Physics Educational Programs accredited clinical medical physics training. SRS and SBRT require significant resources in personnel, specialized technology, and implementation time. A thorough feasibility analysis of resources required to achieve the clinical and technical goals must be performed and discussed with all personnel, including medical center administration. Because various disease sites may have different clinical and technical requirements, feasibility and planning discussions are needed prior to undertaking new disease sites. Treatment of SRS/SBRT patients should adhere to established national guidelines. Acceptance and commissioning protocols and tests must be developed to explore in detail every aspect of the individual and integrated systems with the goal of ensuring safe and effective operation. A comprehensive quality assurance program, encompassing all clinical, technical, and patient-specific treatment aspects, must be developed to ensure SRS and SBRT are performed in a safe and effective manner. Patient safety in radiation therapy is everyone's responsibility. Professional organizations, regulators, vendors, and end-users must demonstrate a clear commitment to working closely together to ensure the highest levels of safety and efficacy in stereotactic radiosurgery and stereotactic body radiation therapy.

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The SBRT white paper was reviewed by 8 experts from the field of SBRT. All the comments were reviewed and discussed by the entire task group and appropriate revisions were incorporated in the paper with task group consensus.

We received approximately 22 comments from physicians, physicists, therapists, and representatives from radiation therapy manufacturers. Additionally, we received general and specific comments from the Associ-

ation of Physicists in Medicine, the American Association of Neurological Surgeons, and the Medical Imaging and Technology Alliance.

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Adherence to this white paper will not ensure successful treatment in every situation. Furthermore, this white paper should not be deemed inclusive of all proper methods of care or exclusive of other methods of care reasonably directed to obtaining the same results. The ultimate judgment regarding the propriety of any specific therapy must be made by the physician and the patient in light of all circumstances presented by the individual situation. ASTRO assumes no liability for the information, conclusions, and findings contained in its white papers. In addition, this white paper cannot be assumed to apply to the use of these interventions performed in the context of clinical trials, given that clinical studies are designed to evaluate or validate innovative approaches in a disease for which improved staging and treatment are needed or are being explored.

This white paper was prepared on the basis of information available at the time the task group was conducting its research and discussions on this topic. There may be new developments that are not reflected in this white paper and that may, over time, be a basis for ASTRO to consider revisiting and updating the white paper.

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